AYE TECHENGINE | COURSE GUIDE

Solar PV Panel Construction

Layers, Cells, Electrical Architecture & Manufacturing

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About This Course Guide

This guide is part of the **AYE TechEngine** series — practical, field-grade learning for engineers, technicians and students. It covers the physical construction, materials, and electrical architecture of solar photovoltaic (PV) panels, with diagrams to help you visualise every layer.

1. Introduction

Solar photovoltaic (PV) panels are precision-engineered devices that convert sunlight directly into electricity. A modern panel is not a single material — it is a **laminated stack** of carefully chosen layers, each performing a specific optical, mechanical, or electrical function. Understanding this construction is essential for anyone who installs, troubleshoots, maintains, or designs solar systems.

In this guide we move from the underlying physics (the photovoltaic effect) to the physical layers, the internal wiring, the cell types, and finally the manufacturing line. By the end you should be able to open any PV panel datasheet and recognise every component listed.

Learning Outcomes

After working through this guide you will be able to: (a) name and describe every layer of a PV panel, (b) explain how the p–n junction generates current from photons, (c) compare monocrystalline, polycrystalline and thin-film cells, and (d) outline the manufacturing sequence from silicon wafer to finished module.

2. How Solar PV Works — The Photovoltaic Effect

A solar cell is essentially a **large-area semiconductor diode**. Silicon is doped to create two adjacent regions: an **n-type** layer (extra electrons) and a **p-type** layer (electron holes). Where they meet — the **p–n junction** — an internal electric field forms.

When photons from sunlight strike the cell with enough energy, they knock electrons loose from silicon atoms. The internal field then pushes electrons toward the n-side and holes toward the p-side. Connecting an external load completes the circuit, and current flows — this is direct-current (DC) electricity.

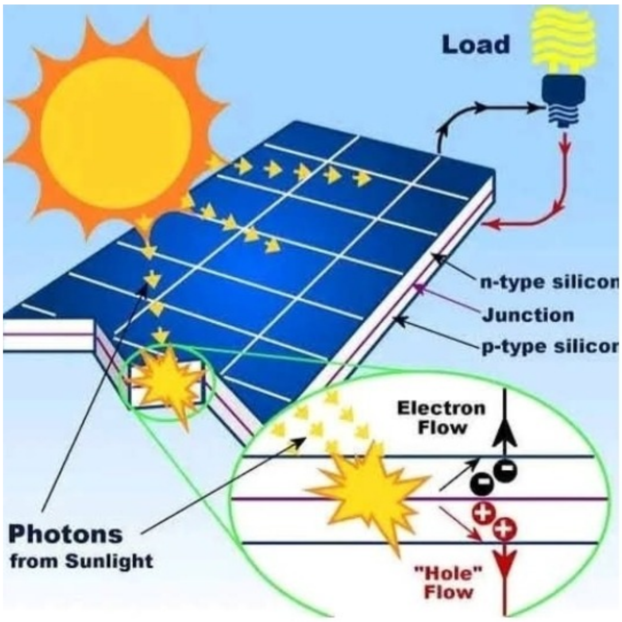

Figure 1 — The photovoltaic effect: photons excite electrons across the p–n junction, creating electron flow (current) and matching hole flow.

Key Physics

Only photons with energy **greater than the silicon band-gap (~1.12 eV)** can free an electron. Lower-energy (infrared) photons pass through, and excess energy from higher-energy photons is lost as heat. This is why no single-junction silicon cell can exceed ~33% theoretical efficiency (the Shockley–Queisser limit).

3. Basic Structure of a Solar PV Panel

A standard crystalline-silicon PV module is built as a laminated sandwich. From front (sun-facing) to back the typical order is:

#	Layer	Function
1	Tempered Glass	Optical window + mechanical protection
2	EVA Encapsulant (front)	Bonds & seals cells, optical coupling
3	Solar Cells (strings)	Photon → electron conversion (DC power)
4	EVA Encapsulant (rear)	Bonds rear side, moisture barrier
5	Backsheet (TPT) or Glass	Electrical insulation, weather barrier
6	Aluminium Frame	Structural rigidity & mounting
7	Junction Box	Cable terminations + bypass diodes

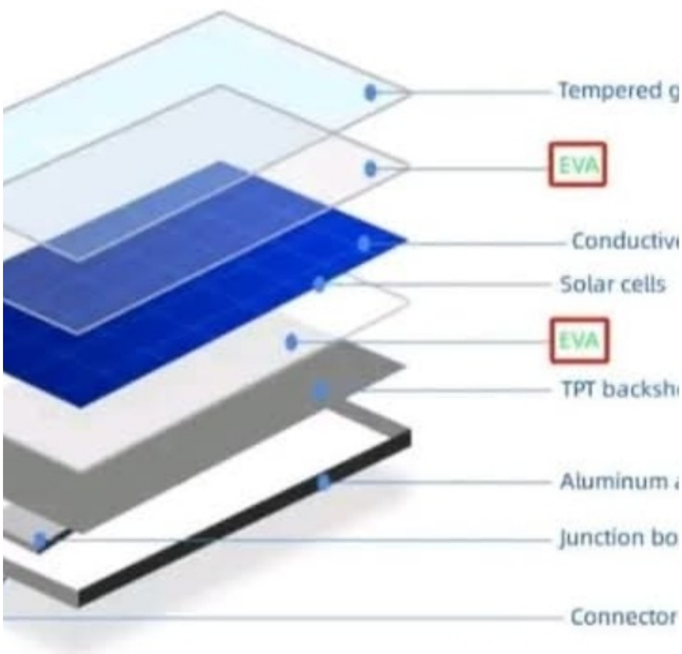

Figure 2 — Exploded view of a standard PV module showing the stacked layers from tempered glass to aluminium frame and junction box.

4. Layer-by-Layer Construction

4.1 Tempered Glass (Front Cover)

Low-iron, high-transmittance tempered glass forms the front surface. It transmits more than 91% of visible light, resists hail impact (typically tested to a 25 mm ice ball at ~23 m/s), and shatters into blunt fragments rather than sharp shards. An anti-reflective coating is often applied to boost transmission.

4.2 EVA Encapsulant

Ethylene-Vinyl Acetate (EVA) is a clear thermoplastic film placed both above and below the cell layer. During lamination it melts, cross-links, and locks the cells in place. EVA provides optical coupling between glass and silicon, electrical insulation, and a moisture seal.

4.3 Solar Cells

Each cell is a thin silicon wafer (typically 156 mm or 166 mm square, ~180 micrometres thick) with a p–n junction, anti-reflective nitride coating, and front/back metallisation. Cells are interconnected by ribbon tabbing into series strings.

4.4 Backsheet (or Rear Glass)

The most common backsheet is a multilayer polymer film known as TPT (Tedlar–PET–Tedlar) — white for residential modules, black for aesthetic 'all-black' designs. It provides electrical insulation and blocks UV and moisture. Bifacial and high-end modules replace the backsheet with a second sheet of glass for longer life.

4.5 Aluminium Frame

Anodised aluminium gives the panel its rigidity, protects the glass edges, and provides standard mounting points for clamps and rails. Frame profiles typically range from 30 mm to 45 mm in depth.

4.6 Junction Box

Mounted on the rear, the junction box terminates the internal cell strings to external solar cables (with MC4 connectors). It houses **bypass diodes** — usually three — which short-circuit shaded sub-strings to prevent hot-spots and power loss.

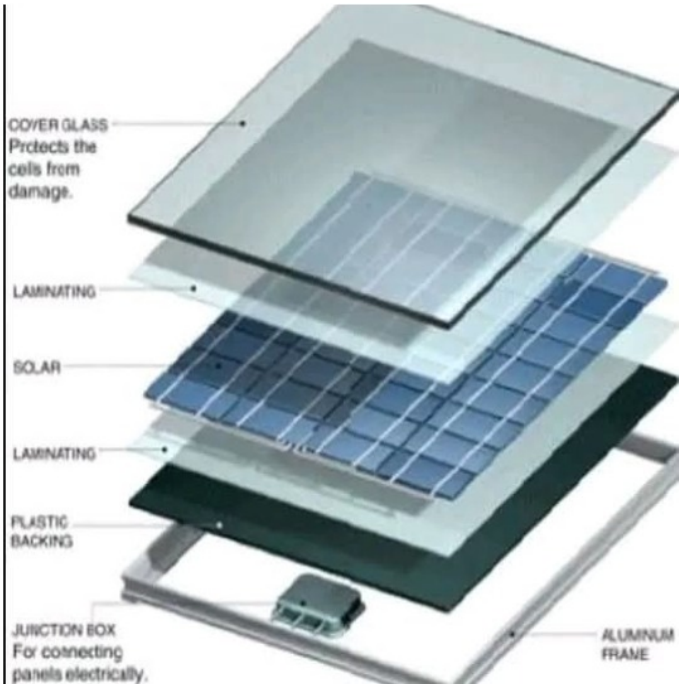

Figure 3 — Cross-section labels: cover glass, EVA laminating layers, solar cells, plastic backing, junction box and aluminium frame.

> **Field Tip**
> When inspecting a panel, always check the **junction box seal** and **frame grounding** first. Most long-term failures (moisture ingress, hot diodes, corrosion) begin at these two locations.

5. Internal Electrical Connections

Inside a finished panel, individual cells are wired into a precise electrical network so that the module delivers a useful voltage and current at its output terminals.

- **Tabbing wires** — thin tinned copper ribbons soldered across the front and back of each cell.
- **Busbars** — main current collectors on each cell (typically 5–12 per cell in modern designs).
- **Series strings** — usually 60, 66 or 72 cells connected in series; each cell adds ~0.6 V, giving panel voltages of roughly 36–44 V open-circuit.
- **Bypass diodes** — three diodes inside the junction box, each protecting one sub-string from shading or cell failure.
- **Output cables** — UV-rated solar cable with polarised MC4 connectors for safe field assembly.

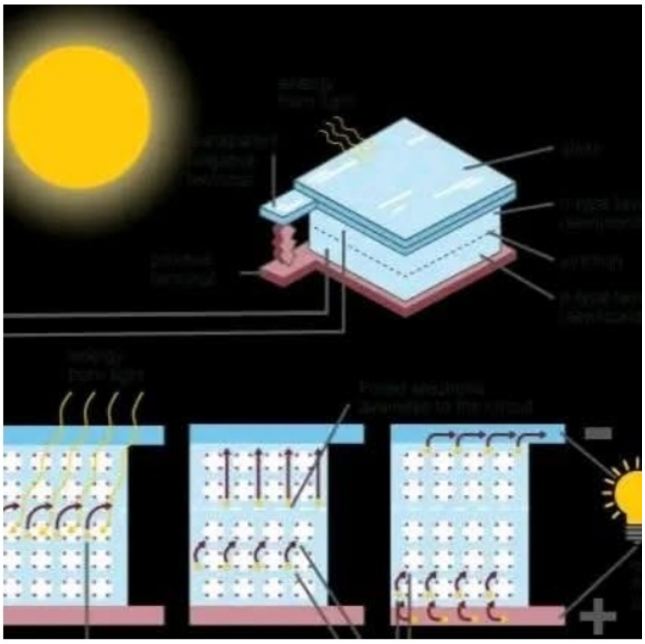

Figure 4 — Cell-level operation and series interconnection: light enters the top conductive layer, generates carriers, and current is collected by busbars wired between cells.

■ Why Bypass Diodes Matter

If even one cell in a 60-cell string is shaded, the entire string can stop producing — and the shaded cell itself may heat dangerously as it is reverse-biased by the others. Bypass diodes give that current a safe alternate path, protecting the cell and recovering most of the panel's output.

6. Types of Solar Cells

Type	Appearance	Typical Efficiency	Best Use Case
Monocrystalline (Mono-Si)	Uniform black	20 – 23 %	Limited rooftop area, premium installs
Polycrystalline (Poly-Si)	Speckled blue	16 – 18 %	Budget commercial / utility farms
PERC (Mono-PERC)	Uniform black	21 – 23 %	Mainstream rooftop & C&I
Bifacial (Glass-Glass)	Transparent rear	20 – 22 %	Ground-mount, carports, trackers
Thin-Film (a-Si/CdTe/CIGS)	Flexible / dark	10 – 15 %	Curved surfaces, BIPV, hot climates

Crystalline silicon (mono + poly + PERC) still dominates the market — more than 95% of global production. Thin-film keeps its niche where weight, flexibility, or low-light performance is critical.

■ Selecting a Panel

When choosing cells, balance **efficiency × area × temperature coefficient × price/W**. A cheaper poly module can outperform a premium mono module if the site is hot and roof space is unlimited.

7. Manufacturing Process (Step-by-Step)

1	**Silicon Purification** Metallurgical-grade silicon is refined via the Siemens process into **polysilicon** of 99.9999%+ purity.
2	**Crystal Growth** Polysilicon is melted and grown into ingots — Czochralski pulling for monocrystalline, directional solidification for polycrystalline.
3	**Wafer Slicing** Ingots are sliced with diamond wire saws into wafers ~170–180 micrometres thick.
4	**Cell Fabrication** Wafers are textured, diffused (to form the p–n junction), coated with anti-reflective silicon nitride, and metallised with screen-printed silver/aluminium contacts.
5	**Cell Testing & Sorting** Each cell is flash-tested under a solar simulator and binned by current to ensure matched strings.
6	**Stringing & Tabbing** Automated stringers solder ribbon onto busbars and interconnect cells into series strings.
7	**Lay-up** Glass → EVA → strings → EVA → backsheet are stacked in a precise sequence.
8	**Lamination** The stack is heated (~150 °C) under vacuum so the EVA melts and cross-links into a sealed, transparent matrix.
9	**Framing & Junction Box** Aluminium frame is bonded with silicone; the junction box is glued, wired and potted on the rear.
10	**Final Test & Class** Modules go through electroluminescence imaging, hi-pot insulation testing, and a final flash test to record nameplate power.

8. Quality Tests & Standards

Before a module ships, it is qualified against international standards. The two most important families are:

- **IEC 61215** — design qualification and type approval (thermal cycling, damp heat, mechanical load, hot-spot endurance).
- **IEC 61730** — safety qualification (insulation, fire rating, partial discharge, accessibility of live parts).
- **UL 1703 / UL 61730** — North-American safety equivalents.
- **Electroluminescence (EL) imaging** — reveals invisible micro-cracks and broken fingers in cells.
- **Flash test** — measures Voc, Isc, Vmp, Imp and Pmax under standard test conditions (STC: 1000 W/m^2, 25 °C, AM1.5).

9. Installation & Maintenance Notes

Knowing how the panel is built directly informs how it should be handled in the field. A few practical reminders:

• Never step on a module — even tempered glass develops micro-cracks under point loads.

• Always carry panels by the frame; do not lift by the junction box or cables.

• Tighten clamps only on the manufacturer-marked zones to avoid frame deformation.

• Inspect the junction box every 1–2 years for moisture, discolouration or burnt diodes.

• Clean panels with soft water and a non-abrasive cloth — avoid pressure washers and hard brushes.

• Use a thermal camera once a year: hot cells and hot diodes are early failure signatures.

Safety First

A PV string under sunlight is always live. Before opening any junction box or removing connectors, cover the module with an opaque sheet, open the DC isolator, and verify zero voltage with a meter rated for DC PV work.

10. Summary & Key Takeaways

A PV panel is more than the silicon you see — it is an optical, mechanical, and electrical system designed to survive 25+ years outdoors. Mastering the layer stack, the cell electronics, and the manufacturing flow gives you the foundation to specify, install, troubleshoot, and even design solar systems with confidence.

Layer / Component	Primary Function
Tempered glass	Light transmission + impact protection
EVA encapsulant	Bonds cells, seals against moisture
Solar cells	Convert photons into DC current
Backsheet / rear glass	Electrical & weather barrier
Aluminium frame	Rigidity & mounting
Junction box + diodes	Termination + shade protection

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